BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Boolean Logic: AND, OR and NOT

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. Why should a corrected answer be retested with a different example?

2. What does an ethical issue concern?

A) What is fair and right
B) Only what is legal
C) Only what is popular
D) Only what is fastest

3. "Open source means there are never any licence conditions or costs." Correct this. Explain why.

4. A computer needs maintenance and data protection. "Utilities are only apps for writing essays." Correct this. What do utilities do?

Today's objectives

Explain the core ideas of boolean logic: and, or and not using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

AND gate
Output 1 only when both inputs are 1.

OR gate
Output 1 when at least one input is 1.

NOT gate
Reverses its single binary input.

Boolean expressions
Symbols and truth tables describe logic gate relationships independently of hardware.

Write one key term and its meaning in your own words:

Do Now	I Do	We Do	You Do	Review & Improve
Recall previous learning	Read new knowledge	Practise with guidance	Work independently	Check my understanding

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New learning - Key ideas

AND gate	Output 1 only when both inputs are 1.
OR gate	Output 1 when at least one input is 1.
NOT gate	Reverses its single binary input.
Boolean expressions	Symbols and truth tables describe logic gate relationships independently of hardware.

I Do - One worked example

For Q = A AND B, complete all four rows of the two-input truth table.

1. List A,B as 00,01,10,11.
2. AND returns 1 only if both are 1.
3. Outputs are 0,0,0,1.
4. Draw a two-input AND symbol matching the rule.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. For Q = A OR B, complete the truth table.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Complete the Boolean test that denies access when authorisation is absent.

```
authorised = false
if ... then
print("Deny")
endif
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. State AND output when A=1, B=0. [1 mark]

Q3. Write OR outputs for inputs 00,01,10,11. [4 marks]

Q4. Explain NOT output for input 0. [2 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. An AND gate outputs 1 only when all its inputs are 1. ☐ T ☐ F
2. Output 1 only when both inputs are 1. ☐ T ☐ F
3. Output 1 when at least one input is 1. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 To check that the underlying method works independently on another case. ☐

Q2 A - What is fair and right ☐

Q3 It permits access to source code and modification or redistribution under its conditions. ☐

Q4 Utilities maintain or protect a computer, adding tools beyond the OS. Writing an essay is an application task. ☐

Code First - Model

```
authorised = false
if NOT authorised then
print("Deny")
endif
```

Output: Deny

We Do - Check the model

1. Outputs 0,1,1,1. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 0. ☐

Q3 0,1,1,1. ☐

Q4 NOT inverts the bit; output 1. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer:

Do Now	I Do	We Do	You Do	Review & Improve
Recall previous learning	Read new knowledge	Practise with guidance	Work independently	**Check my understanding**